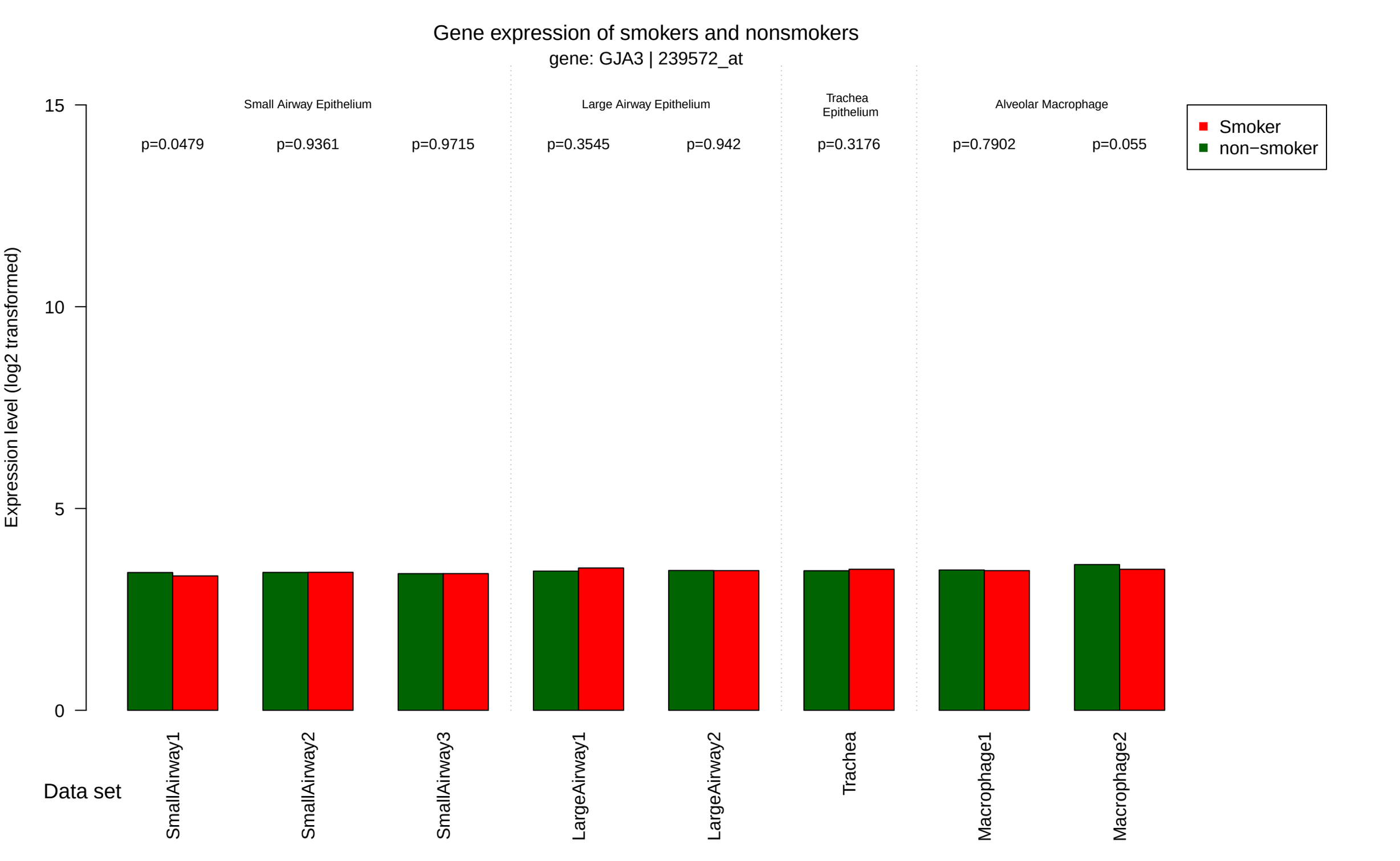
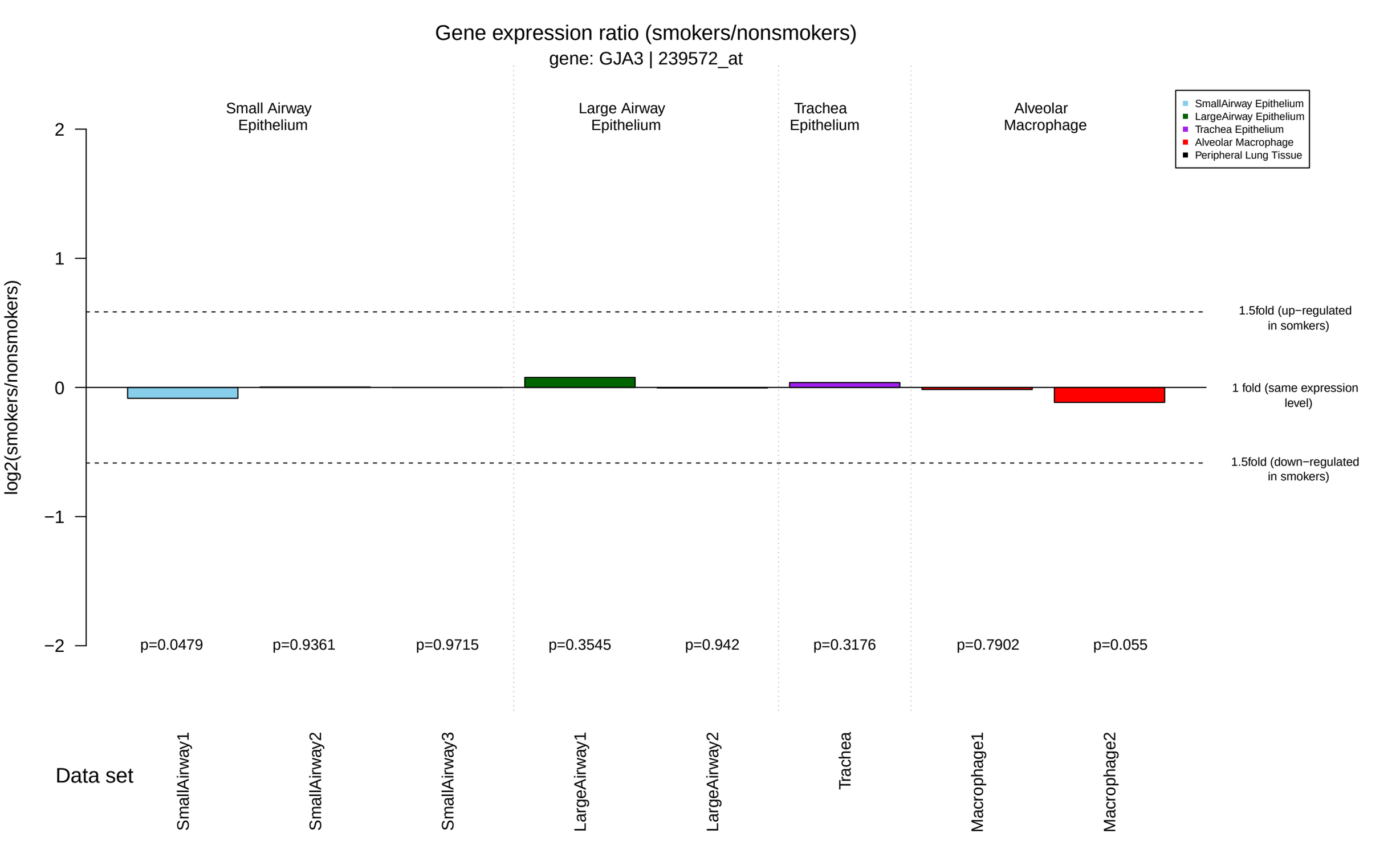
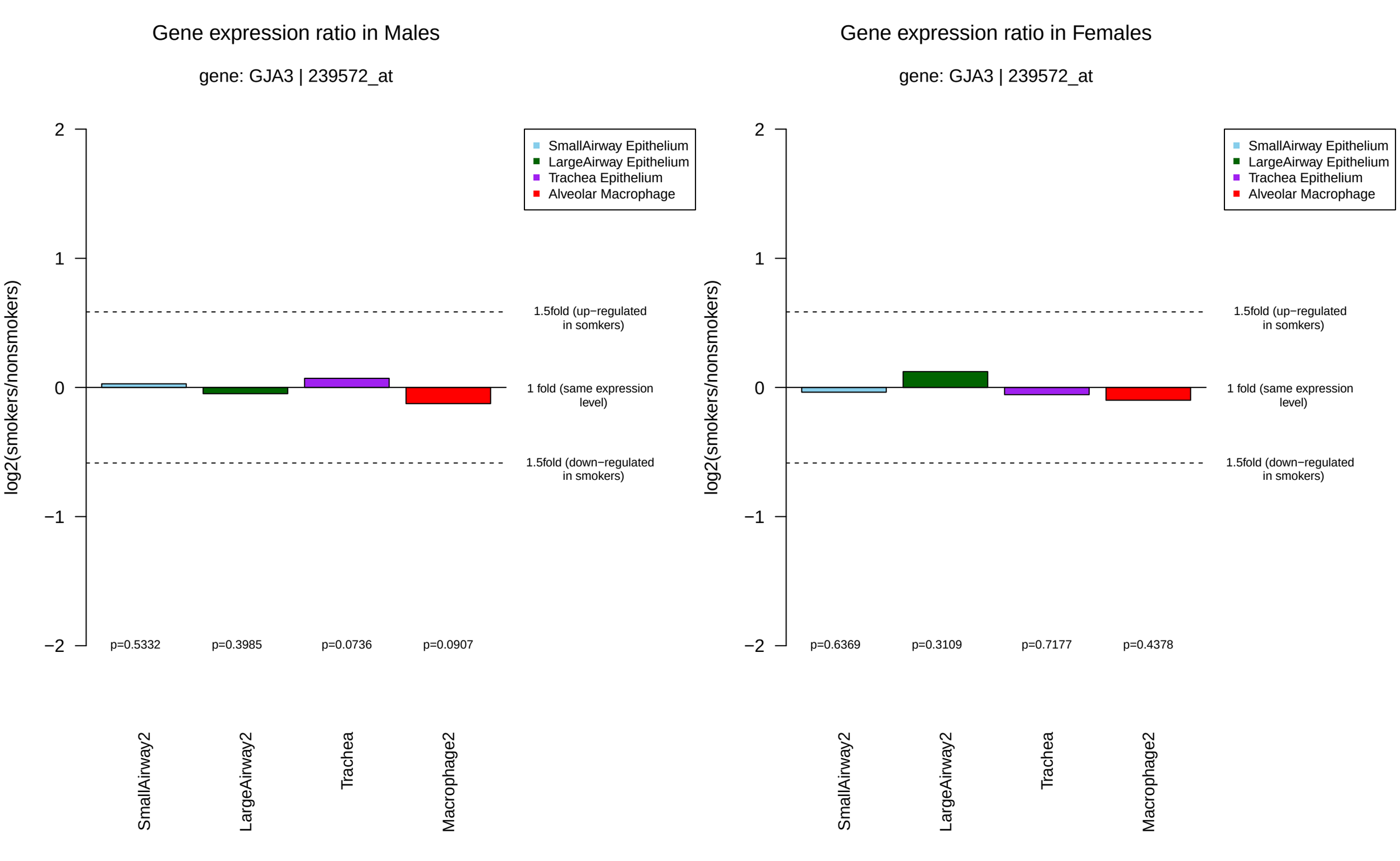
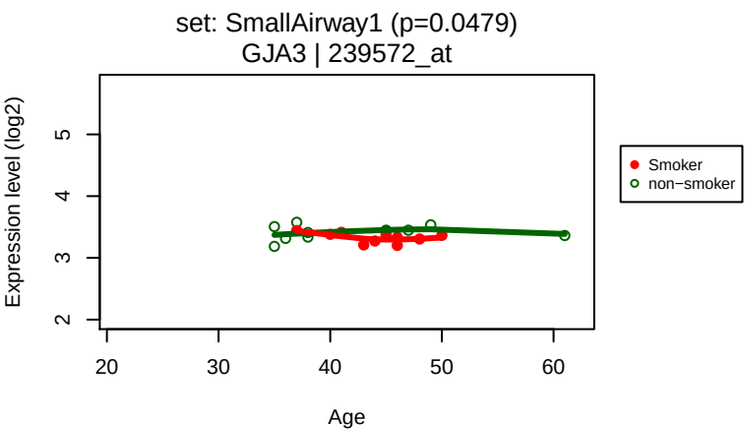
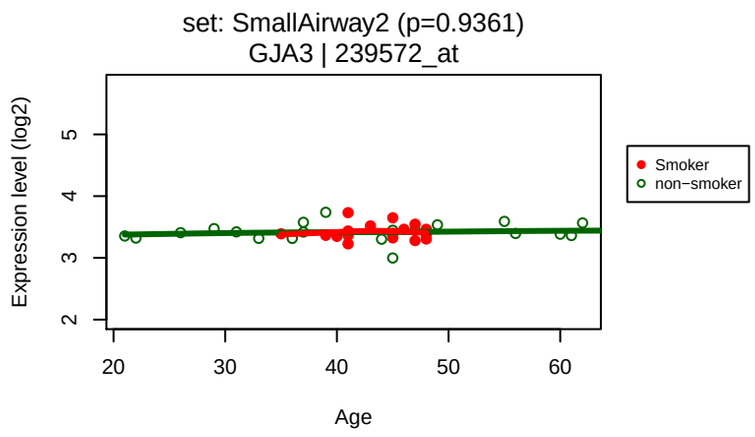
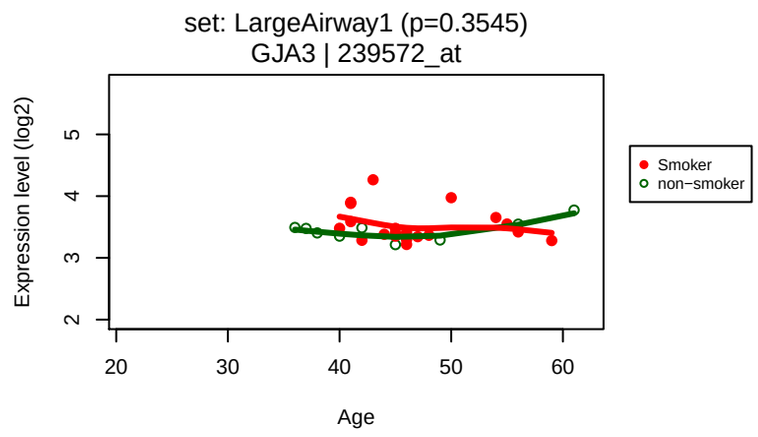

Gene expression of smokers and nonsmokers
gene: GJA3 | 239572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0479
p=0.9361
p=0.9715
p=0.3545
p=0.942
p=0.3176
p=0.7902
p=0.055
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

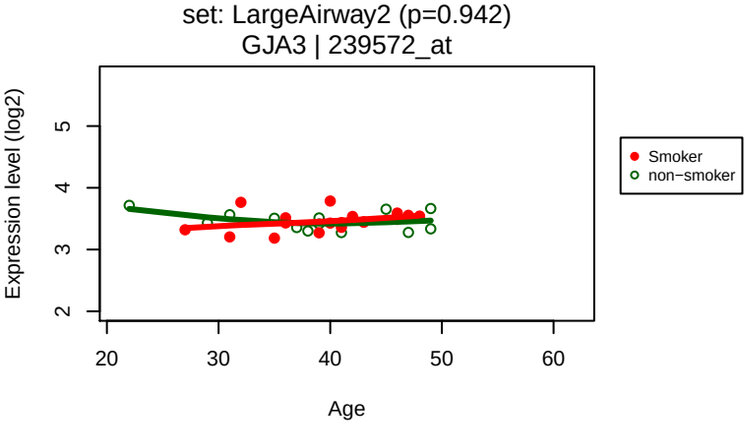

Gene expression ratio (smokers/nonsmokers)
gene: GJA3 | 239572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0479
p=0.9361
p=0.9715
p=0.3545
p=0.942
p=0.3176
p=0.7902
p=0.055
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

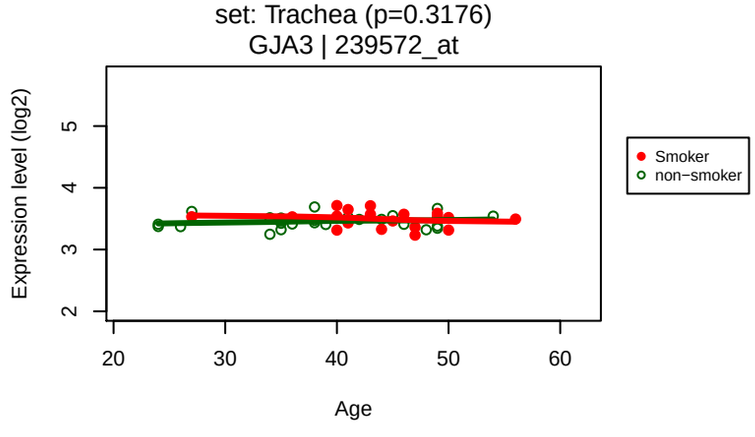

Gene expression ratio in Males
gene: GJA3 | 239572_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5332
p=0.3985
p=0.0736
p=0.0907
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GJA3 | 239572_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6369
p=0.3109
p=0.7177
p=0.4378
SmallAirway2
LargeAirway2
Trachea
Macrophage2

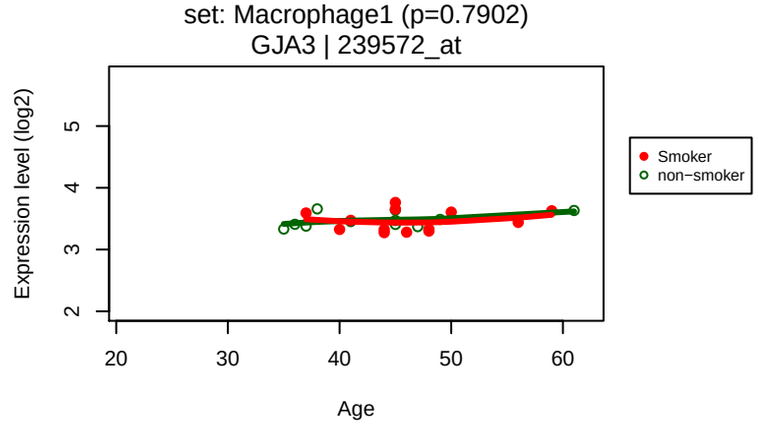

set: SmallAirway1 (p=0.0479)
GJA3 | 239572_at
Expression level (log2)
Age
Smoker
non−smoker

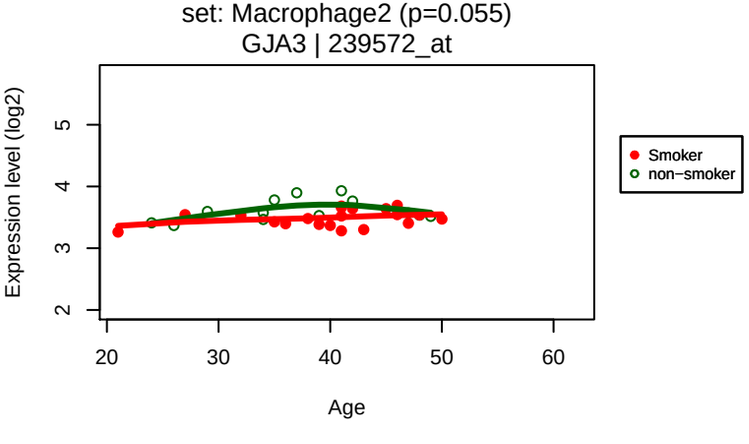

set: SmallAirway2 (p=0.9361)
GJA3 | 239572_at
Expression level (log2)
Age
Smoker
non−smoker

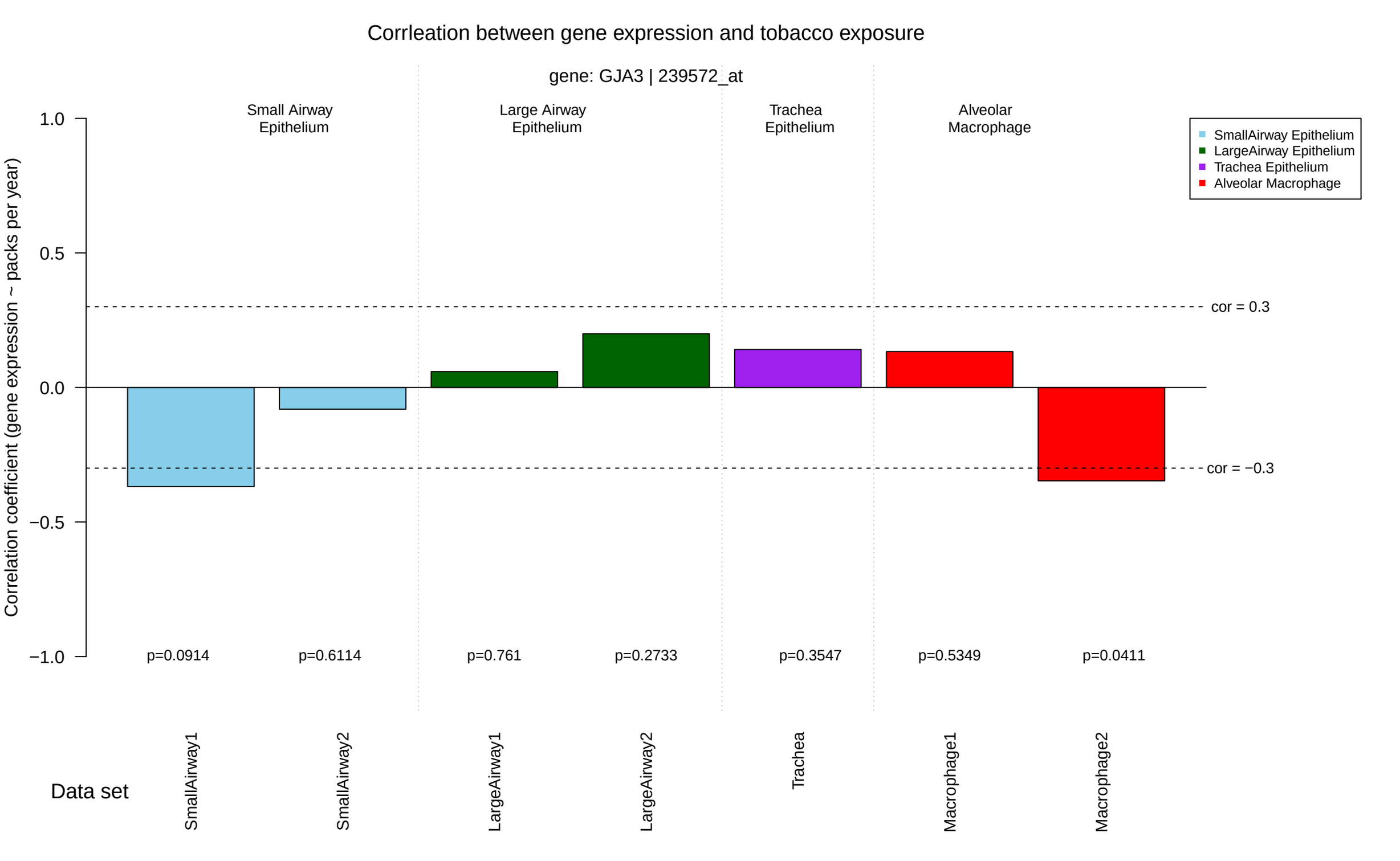

set: LargeAirway1 (p=0.3545)
GJA3 | 239572_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.942)
GJA3 | 239572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3176)
GJA3 | 239572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7902)
GJA3 | 239572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.055)
GJA3 | 239572_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GJA3 | 239572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0914
p=0.6114
p=0.761
p=0.2733
p=0.3547
p=0.5349
p=0.0411
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2